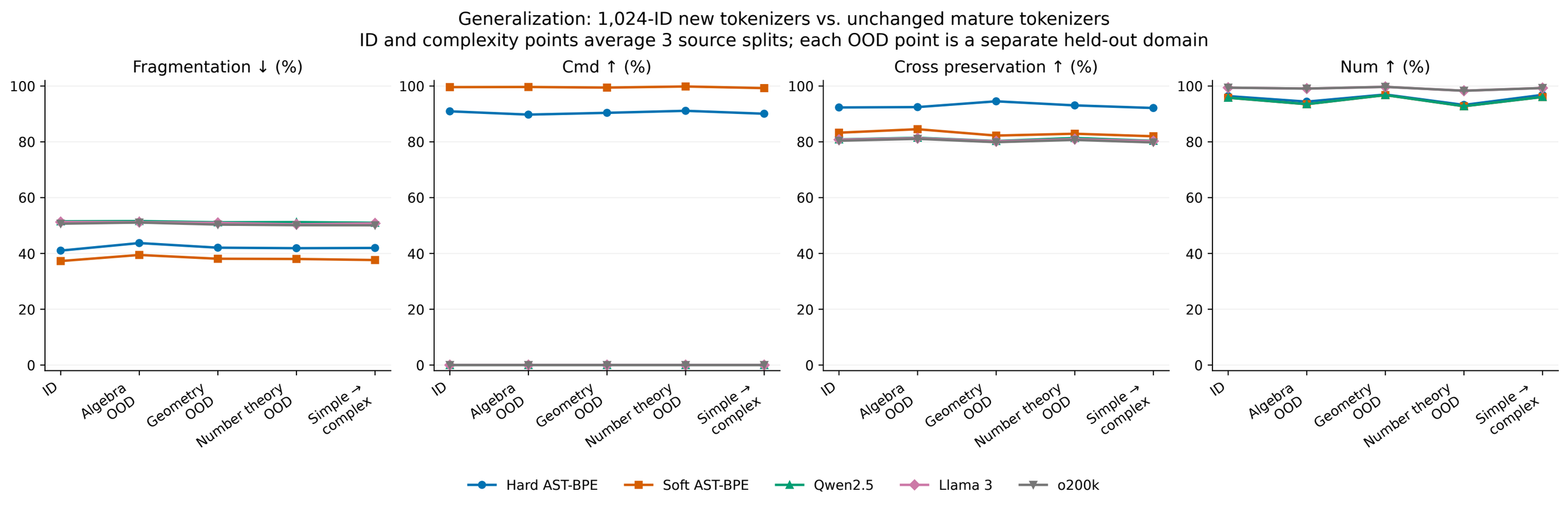
Generalization: 1,024-ID new tokenizers vs. unchanged mature tokenizers
ID and complexity points average 3 source splits; each OOD point is a separate held-out domain
Fragmentation ↓ (%)
Cmd ↑ (%)
Cross preservation ↑ (%)
Num ↑ (%)
100
80
60
40
20
0
ID
Algebra OOD
Geometry OOD
Number theory OOD
Simple → complex
Hard AST-BPE
Soft AST-BPE
Qwen2.5
Llama 3
o200k